

Excluding the first four odd-degree parity-bound candidates

Mathematical proof independently reviewed; not formally verified

30 September 2026

Write $v = v_2$. We use the parity-count and resonance conclusions proved in `../independent_directions/optimized_two_adic_filter.tex`. For odd degree r , if a rational equal-weight rule has $2^{r-1} \leq K < 2^r$, and $N = 2^a$ is the least power of two making all nodes $y_i = Nx_i$ integral in $\mathbb{Z}_2$, then exactly 2^{r-1} normalized nodes are odd, $N \mid r + 1$, and

$$v(K) = (r + 1)/N - 1. \quad (1)$$

The case where all original nodes are integral forces $2^r \mid K$.

Theorem 1. *For $r = 2^a - 1$ with $a \geq 3$, no rational equal-weight uniform-interval rule of degree r has $K = 2^{r-1} + 1$ nodes. Consequently*

$$R_r \geq 2^{r-1} + 2.$$

In particular $R_7 \geq 66$. The nodes need not be in $[0, 1]$ for this lower bound.

Proof. Suppose such a rule exists. Its K is odd, so (1) forces $N = r + 1 = 2^a$. Exactly 2^{r-1} normalized nodes are odd, and there is one even node z . Put

$$k = (r - 1)/2 = 2^{a-1} - 1, \quad f(y) = y(y^2 - 1)^k, \quad g(y) = (y^2 - 1)^k.$$

Both polynomials have degree at most r . For odd $y \in \mathbb{Z}_2$, $y^2 - 1 \in 8\mathbb{Z}_2$, so $f(y), g(y) \in 2^{3k}\mathbb{Z}_2$.

The integral of $f(Nx)$ is

$$A = \sum_{j=0}^k \binom{k}{j} (-1)^{k-j} \frac{N^{2j+1}}{2j+2}.$$

The $j = 0$ term has valuation $a - 1$. For $j \geq 1$, the valuation is at least $(2j+1)a - 1 - v(j+1) > a - 1$. Thus $v(KA) = a - 1$. Exactness gives $f(z) \equiv KA \pmod{2^{3k}}$, and $3k > a - 1$. Since z is even, $v(f(z)) = v(z)$, proving

$$v(z) = a - 1. \quad (2)$$

As k is odd, expansion of $g(z) - (-1)^k$ has leading nonzero term $k(-1)^{k-1}z^2$. All later terms have greater valuation, and (2) yields

$$v(g(z) - (-1)^k) = 2a - 2. \quad (3)$$

On the other hand, writing $B = K \int_0^1 g(Nx) dx$, we have

$$B - (-1)^k = (-1)^k(K - 1) + K \sum_{j=1}^k \binom{k}{j} (-1)^{k-j} \frac{N^{2j}}{2j+1}.$$

Here $v(K - 1) = r - 1 = 2^a - 2 \geq 2a$, and every summand in the sum has valuation at least $2a$. Therefore $v(B - (-1)^k) \geq 2a$. Exactness for g gives $g(z) \equiv B \pmod{2^{3k}}$; since $3k \geq 2a$, this contradicts (3). $\square$

Theorem 2 (Four consecutive exclusions). *For $r = 2^a - 1$, $a \geq 3$, one has*

$$R_r \geq 2^{r-1} + 5.$$

In particular $R_7 \geq 69$.

Proof. The preceding theorem excludes the first candidate. Suppose now that $K = U + s$, where $U = 2^{r-1}$ and $s \in \{2, 3, 4\}$. The resonance condition (1) immediately excludes $s = 4$: it would give $v(K) = 2$ and $(r + 1)/N = 3$, impossible because $r + 1$ and N are powers of two.

Retain $k = (r - 1)/2$, $f(y) = y(y^2 - 1)^k$, and $g(y) = (y^2 - 1)^k$. For $s = 2$, (1) gives $N = 2^{a-1}$ and two even normalized nodes; for $s = 3$, it gives $N = 2^a$ and three even normalized nodes. In both cases the same uniquely minimal linear term in the integral of $f(Nx)$ gives

$$v\left(K \int_0^1 f(Nx) dx\right) = a - 1. \quad (4)$$

All odd-node contributions are in $2^{3k}\mathbb{Z}_2$.

We use an elementary valuation observation. For even z put

$$h(z) = g(z) + 1 = z^2 Q(z^2), \quad Q(0) = k \text{ odd},$$

where Q has integer coefficients. For all z of a fixed finite valuation $b \geq 1$, the odd quantity $h(z)/2^{2b}$ has the same residue $Q(2^{2b})$ modulo eight: write $z = 2^b u$ and use $u^2 \equiv 1 \pmod{8}$. For a sum of two or three such $h(z)$, let b be the smallest valuation of the nonzero z 's. If one or three terms attain b , the sum has valuation $2b$. If exactly two terms attain b , their normalized sum is two or six modulo eight. A possible third term contributes zero or four modulo eight, so the sum has valuation $2b + 1$.

For $s = 2$, expansion of the integral gives

$$K \int_0^1 g(Nx) dx + 2 = -U + K \sum_{j=1}^k \binom{k}{j} (-1)^{k-j} \frac{N^{2j}}{2j+1}.$$

Its $j = 1$ term has valuation $1 + 2(a - 1) = 2a - 1$; all other terms, including U , have greater valuation. Exactness and $3k > 2a - 1$ show that the sum of the two $h(z)$ has valuation $2a - 1$. The observation forces both even nodes to have valuation $a - 1$. Their f values, divided by 2^{a-1} , are both odd, so their sum has valuation at least a , contradicting (4).

For $s = 3$ and $a \geq 4$, the analogous integral plus three has exact valuation $2a$: its $j = 1$ term is uniquely minimal, since $v(U) = 2^a - 2 > 2a$. The observation forces every even node to have valuation at least a . Their f values then sum to an element of $2^a\mathbb{Z}_2$, again contradicting (4). For the remaining case $a = 3, s = 3$, the first two terms combine as $-64 + 67 \cdot 64 = 64 \cdot 66$, with exact valuation seven; all later terms have valuation at least twelve. Hence the sum of the three $h(z)$ has valuation seven. The observation forces exactly two nodes of valuation three and the remaining node of valuation at least four. Their f values sum with valuation at least four, contradicting the required valuation two in (4). $\square$

Review and origin of the strengthening. The initial one-node exclusion and the degree-seven congruences below were checked independently by two parallel reviewers. One reviewer then supplied the two-node exclusion and the general $s = 2, 3, 4$ extension; the three-node degree-seven case was independently derived. The resulting four-exclusion proof was checked separately case by case. This is mathematical peer review, not formal proof verification.

The degree-seven calculation directly. If $K = 65$, the normalization is $N = 8$, with 64 odd nodes and one even node z . The polynomial $y(y^2 - 1)^3$ takes values in $512\mathbb{Z}_2$ on odd nodes, whereas

$$65 \int_0^1 8x((8x)^2 - 1)^3 dx = 65(2^{18} - 2^{14} + 384 - 4)$$

has valuation two. Thus $v(z) = 2$. But $(y^2 - 1)^2$ takes values in $64\mathbb{Z}_2$ on odd nodes and

$$65 \int_0^1 ((8x)^2 - 1)^2 dx = 65 \left(\frac{4096}{5} - \frac{128}{3} + 1 \right) \equiv 1 \pmod{64}.$$

For $v(z) = 2$, $(z^2 - 1)^2 \equiv 33 \pmod{64}$, a contradiction.

A further obstruction for symmetric degree-seven rules

The general lower bound is $R_7 \geq 69$. The next candidate cannot be attained by a rule symmetric about $1/2$.

Theorem 3. *No symmetric rational equal-weight degree-seven rule has 69 nodes.*

Proof. The resonance condition gives least normalization $N = 8$, with 64 odd nodes. Center the normalized coordinates as $u = 8x - 4$. Symmetry makes the five even coordinates a multiset $0, v, -v, w, -w$, where $v, w \in 2\mathbb{Z}_2$; initially either may be zero. Put $g(u) = (u^2 - 1)^3$ and $h(u) = g(u) + 1$. The odd coordinates contribute 64 modulo 512 to the sum of h . On the other hand,

$$69 \int_{-1}^1 h(4t) \frac{dt}{2} - 64 = 69 \left(\frac{4096}{7} - \frac{768}{5} + 16 \right) - 64 = \frac{16 \cdot 67411}{35},$$

which has valuation 4. Thus $v_2(h(v) + h(w)) = 3$. The fixed-valuation residue argument used above shows that two nonzero even inputs give valuation $2b$ if their valuations differ, and $2b + 1$ if both have valuation b . Consequently $v_2(v) = v_2(w) = 1$.

It follows that $v^2, w^2 \equiv 4 \pmod{32}$, hence $g(v), g(w) \equiv 27 \pmod{32}$. The even-node contribution to g is therefore

$$g(0) + 2g(v) + 2g(w) \equiv -1 + 4 \cdot 27 = 43 \pmod{64}.$$

The odd-node contribution vanishes modulo 512, whereas exactness requires

$$69 \int_{-1}^1 g(4t) \frac{dt}{2} = 69 \left(\frac{4096}{7} - \frac{768}{5} + 15 \right) \equiv 11 \pmod{64},$$

a contradiction. This argument does not exclude nonsymmetric 69-node rules. $\square$

An explicit seventy-node rule

There is a symmetric rational degree-seven rule with 70 nodes. Take $x = (18 \pm s)/36$, with the following multiplicities for each sign:

s	1	3	4	5	7	10	11	12	13	15	17
c_s	3	3	1	3	9	1	2	1	4	4	4.

Indeed $\sum c_s = 35$, and

$$\sum c_s s^2 = 3780 = \frac{35 \cdot 18^2}{3}, \quad \sum c_s s^4 = 734832 = \frac{35 \cdot 18^4}{5}, \quad \sum c_s s^6 = 170061120 = \frac{35 \cdot 18^6}{7}.$$

Symmetry supplies the odd centered moments. The rule therefore has all required moments through degree seven. Repeated nodes are allowed in an equal-weight rule. The standalone verifier `verify_degree7_seventy.py` checks all eight ordinary moments with exact rational arithmetic.

Consequently

$$\boxed{69 \leq R_7 \leq 70},$$

and the minimum number of nodes among symmetric rational degree-seven rules is exactly 70. Whether a nonsymmetric 69-node rule exists remains open in these notes.